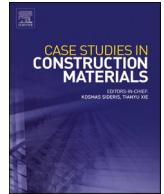




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Enhancing asphalt mix design with transfer learning and hybrid artificial intelligence

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ABSTRACT

Asphalt content (AC) in hot mix asphalt is a critical factor influencing pavement durability and performance. Traditional determination methods are labor-intensive, costly, and prone to human error, limiting scalability. Artificial intelligence (AI) and machine learning (ML) offer efficient alternatives that improve accuracy and reduce manual effort, but they typically require large training datasets. This study addresses that challenge by proposing a hybrid transfer learning model (RF-DNN-T) that leverages a comprehensive literature-based dataset of 848 mix designs for pre-training, followed by fine-tuning with only 96 target-specific samples. The framework combines the feature-selection capabilities of Random Forest (RF) with the nonlinear pattern recognition strength of Deep Neural Networks (DNN) to enhance predictive efficiency and reliability. By transferring knowledge from multiple laboratories, the model minimizes experimental effort while maintaining high accuracy. The RF-DNN-T model is benchmarked against conventional ML approaches using SHapley Additive exPlanations (SHAP) for interpretability. Results show the model achieves the lowest testing errors (MAE = 0.08, MSE = 0.02, MAPE = 1.61) and the highest R^2 (0.97), indicating excellent accuracy, generalization, and a strong correlation between predictions and experimental data. SHAP analysis identifies voids as the most influential feature, contributing 32 % to prediction impact. This work demonstrates that the RF-DNN-T model can accurately predict AC with minimal laboratory data, significantly reducing experimental workload while preserving precision in asphalt mix design.

1. Introduction

Accurate prediction of bitumen content is crucial in hot mix asphalt design because it directly affects structural integrity, durability, and workability [1]. By ensuring proper compaction and air void distribution, it minimizes distress such as rutting, fatigue cracking, and moisture damage [2,3]. This prediction relies on key volumetric and mechanical properties, including Marshall Stability (MS), Flow, Air Voids (Va), Voids in Mineral Aggregate (VMA), and Voids Filled with Asphalt (VFA), which collectively determine resistance to deformation, fatigue, and environmental stressors [4]. If asphalt mixtures fail to meet these criteria, they are prone to thermal and fatigue cracking, moisture-induced stripping, and permanent deformation, accelerating deterioration and jeopardizing road safety [5,

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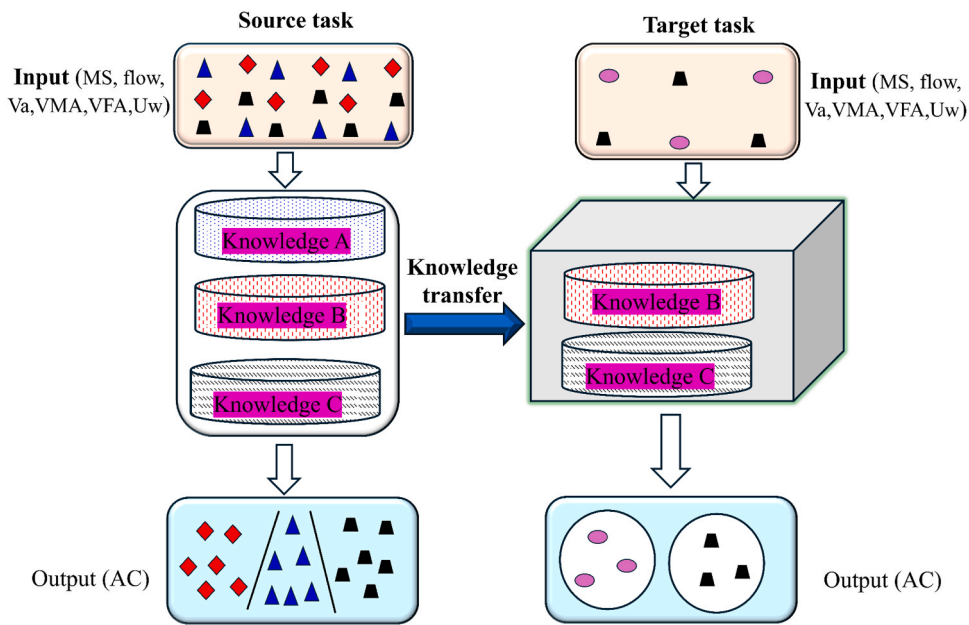


Fig. 1. Illustration of transfer learning in asphalt content prediction.

6]. Such failures shorten pavement lifespan, raise maintenance costs, and increase safety risks, highlighting the importance of advanced mix design methods [7,8]. Therefore, precise characterization of asphalt mixtures based on composition is essential for optimizing performance in the design process [9].

The Marshall and modified Marshall methods, based on aggregate gradation and bitumen content, remain the dominant empirical approaches for determining asphalt content in pavement construction. Lacking explicit mathematical formulations, they rely on trial-and-error [10], requiring extensive laboratory testing and skilled labor, which increases cost and delays projects [9]. Single-factor experimental designs further limit accuracy by neglecting interactions among mixture constituents [11]. These limitations have driven the adoption of advanced, data-driven optimization techniques. Researchers have explored numerical and mathematical models to improve asphalt content prediction and simulate mechanical behavior with greater efficiency and accuracy [12]. Approaches such as statistical modeling, mechanistic-empirical methods, and predictive analytics have been developed to overcome the shortcomings of traditional mix design [13–16]. Early studies proposed polynomial models for estimating optimal bitumen content, while Zhang et al. [17] applied statistical methods to streamline experimentation. Yildirim et al. [18] employed Central Composite Design to enhance prediction accuracy in Marshall design, and Zhang et al. [19] introduced a theoretical model for rubberized asphalt mix design, further advancing estimation methodologies.

Despite advances in predictive modeling, regression-based approaches struggle to capture complex volumetric interactions in asphalt mixtures, leading to structural bias and poor representation of nonlinear viscoelastic behavior [6]. Many rely on empirical assumptions and simplified temperature functions requiring costly laboratory validation, while mechanistic-empirical models remain highly sensitive to calibration dataset quality, limiting reliability across materials and conditions [20]. Given rising asphalt production and maintenance costs, integrating next-generation computational intelligence is now essential for precision, efficiency, and sustainability in mix design and quality control [21].

The rising demand for time-efficient and scientifically robust mix design has accelerated the adoption of artificial intelligence (AI) in pavement engineering. AI has been applied to performance assessment [22–24], pavement distress detection [25,26], Marshall parameter prediction [27], flow number estimation [28], road roughness evaluation [29], International Roughness Index estimation [30,31], asphalt mix optimization [32–34], compressive strength prediction [35–37], sustainable concrete design [38], rheological property prediction [39], and maintenance planning [40,41]. These approaches address key limitations of regression-based and mechanistic-empirical models, offering greater accuracy and efficiency. Numerous studies highlight AI's potential: Azarhoosh and Pourasmaeil [6] used Genetic Programming to model Marshall parameters; Zhang et al. [34] applied Support Vector Machines to enhance asphalt pavement design; Althoeay et al. [7] reported DT-Bagging achieved the highest prediction accuracy; Wahab et al. [42] found CatBoost outperformed empirical models in predicting fiber-reinforced polymer-concrete bond strength; Shah et al. [43] used ANNs to model Marshall stability under varying aggregate and temperature conditions; and Chen et al. [44] implemented a hybrid RF-NSGA-II model to optimize high-performance concrete, improving durability and cost efficiency.

Huang et al. [45] employed LightGBM to predict self-healing and durability of high-performance concrete. Wang et al. [46] created a multi-objective ML model for recycled aggregate concrete, identifying fly ash as the most effective SCM. Tuken et al. [47] applied CatBoost to predict load-deflection in steel fiber-reinforced concrete. Wang et al. [48] proposed a Monte Carlo-based framework for marine concrete service life prediction. Wang et al. [49] introduced a Bayesian-optimized CTGAN model to address data scarcity in

low-carbon concrete design. Al-Sabaei et al. [39] combined response surface methodology and ML to predict the rheology of modified asphalt binders with high accuracy ($R^2 = 0.99$). Wang et al. [38] reviewed AI-driven methods for predicting and optimizing sustainable concrete, emphasizing data complexity, model transparency, and the need for robust databases. Milad et al. demonstrated that Artificial Neural Networks (ANNs) outperform traditional methods for predicting bitumen rheology [50] and successfully applied ML to forecast pavement maintenance with high accuracy [41]. Similarly, Tangga et al. [27] utilized ensemble models like AdaBoost to predict asphalt pavement characteristics with exceptional R^2 values up to 99.8 %. Further supporting this trend, Anwar et al. [51] evaluated a range of ML models for predicting the compressive strength of fly ash-based geopolymers, finding that ensemble methods and ANNs yielded the highest accuracy. Beyond material properties, Araba et al. [52] reviewed computational methods for project cost estimation, emphasizing the effectiveness of Estimate at Completion techniques for managing risk, while Abdolrasol et al. [53] explored how integrating optimization algorithms with ANNs enhances performance in fields like energy management.

Collectively, these AI-driven methodologies offer a data-driven, cost-effective alternative to traditional empirical approaches, significantly reducing labor and material costs while enhancing predictive performance. Despite these advances, traditional ML models continue to face key limitations, including limited data availability, high computational requirements, susceptibility to overfitting, and poor generalizability [54–57]. These models often demand large, labeled datasets, extensive hyperparameter tuning, and significant computational resources, making them less feasible for use in resource-constrained settings. Their tendency to overfit training data further undermines predictive reliability, especially in real-world scenarios where obtaining high-quality data is both costly and challenging [54].

To address the limitations of traditional ML, transfer learning has emerged as a promising alternative. By leveraging pre-trained models from related domains, it improves predictive accuracy, generalization, and computational efficiency in data-scarce contexts [55,58]. Unlike conventional ML, it mitigates overfitting, accelerates convergence, and reduces dependence on large labeled datasets. Although it adds some complexity, transfer learning offers a scalable, robust, and data-efficient approach for asphalt mix design and pavement performance modeling [5,59]. Fundamentally, it transfers knowledge from a source task to a related target task, enhancing learning efficacy where data are limited. Fig. 1 illustrates its application in asphalt content prediction.

Transfer learning is gaining momentum in civil engineering due to its potential to overcome data scarcity and improve model performance [60–63]. Several researchers have explored its application across diverse subfields. For instance, Moussa and Owais [55] developed a transfer learning-based model for predicting HMA dynamic modulus, significantly reducing the need for laboratory testing. Li et al. [64] introduced an improved TrAdaBoost algorithm for pavement performance forecasting, while Liu et al. [65] applied a CNN-based transfer learning model for pavement vibration analysis, achieving high accuracy. Zhao et al. [66] used a hybrid XGBoost-DNN model for driver speed estimation, leveraging transfer learning to boost performance. Similarly, Hu et al. [67] applied transfer learning for seismic response prediction under data-limited conditions, and Bellagarda et al. [68] used it to enhance temperature forecasting in smart buildings.

Martinez-Ríos et al. [69] applied transfer learning to pavement crack detection, with SqueezeNet achieving the highest sensitivity among tested models. Nguyen et al. [70] developed a transfer learning based model for predicting concrete compressive strength, while Wang and Gan [71] used transfer learning for automated building inspection, improving defect detection efficiency. Fang et al. [72] proposed a hybrid LSTM-DANN model for cross-building energy consumption prediction, effectively capturing temporal and domain-invariant features. In indoor environments, Somu et al. [73] implemented a CNN-LSTM architecture to predict thermal comfort under dynamic conditions. Hamida et al. [74] employed a RF-DNN hybrid for skin disease classification, and Elbeltagi et al. [75] enhanced evapotranspiration prediction using a DNN integrated with random forest. Nourani et al. [76] developed a hybrid CNN-LSTM model incorporating data fusion and transfer learning to accurately forecast solar irradiation in limited-data scenarios.

These advancements underscore the potential of transfer learning to enhance model generalization, reduce computational demands, and address data scarcity across engineering applications. Nevertheless, its use in transportation particularly asphalt mix design remains limited, presenting a critical research gap. Despite progress in AI applications, three major challenges persist: limited generalizability under data scarcity, dependence on costly and labor-intensive laboratory testing, and underutilization of transfer learning. To address these issues, this study introduces a domain-adaptation transfer learning approach through a novel RF-DNN-T model, which transfers knowledge from a large source dataset to a smaller target dataset, substantially reducing laboratory testing requirements. This approach builds on the field's evolution from statistical models to advanced machine and deep learning techniques, extending the recent trend of ensemble and hybrid methods by combining the robust feature selection of RF with the deep feature-learning capabilities of DNN.

This study addresses the challenge of accurately predicting AC for cost-effective pavement design under complex, data-scarce conditions. A novel hybrid deep transfer learning model, RF-DNN, is developed by integrating the robust feature selection of Random Forest with the enhanced generalization of a DNN. Pre-trained RF-DNN models are fine-tuned with limited laboratory data, substantially reducing the need for costly testing. Model performance is rigorously evaluated through statistical analysis, demonstrating superior accuracy and robustness. Interpretability is enhanced using the SHAP method to quantify feature influence. The proposed framework provides a scalable, AI-driven solution for asphalt mix optimization, reducing laboratory effort and overall costs.

2. Materials and method

2.1. Data set

The dataset is a critical component in the technical implementation of ML, as its quality, diversity, and consistency directly affect model performance during both training and prediction phases. This study employs a comprehensive dataset divided into two main



Fig. 2. Ethiopian Iron and Steel Factory: (a) Raw materials, including stacks of processed steel and iron ingots, used in the production of steel and iron; (b) Steel slag, a solid waste byproduct generated during the steel manufacturing process, typically consisting of impurities and residual materials.

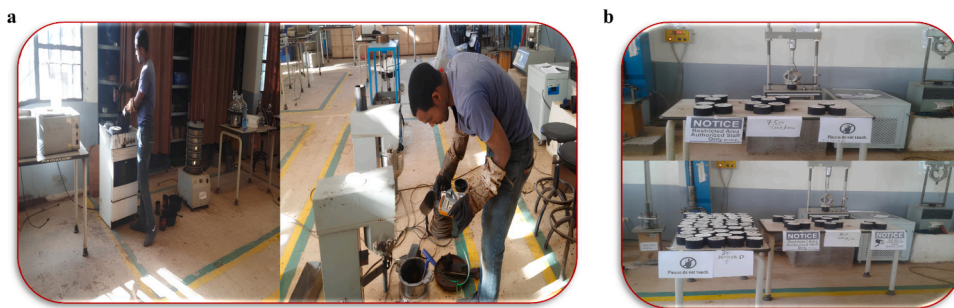


Fig. 3. Laboratory procedures for Marshall mix design: (a) Preparation of the mix, including material selection, heating, and mixing aggregates with asphalt binder; (b) Samples prepared for compaction using a Marshall hammer.

categories: the source dataset and the target dataset. The source dataset, which forms the basis for transfer learning, consists of an extensive collection of volumetric and mechanical properties of asphalt mixtures such as Marshall Stability, Flow, Air Voids, Voids in Mineral Aggregate (VMA), Voids Filled with Asphalt (VFA), and mix density compiled from a broad range of literature and open sources. This rich dataset enables the development of a pre-trained ML model, improving its capability to identify patterns and generalize across different asphalt mixture properties.

The target dataset was developed through controlled laboratory experiments to evaluate the use of steel slag as a filler material in hot mix asphalt (HMA), with varying contents from 0 % to 10 %. Steel slag, a by-product from the Ethiopian Iron and Steel Factory, was used as a sustainable replacement for stone dust (passing the #200 sieve) to minimize environmental impact, as illustrated in Fig. 2. Materials used in the mix included coarse and fine aggregates from IFH Engineering PLC's Meshenti quarry, 80/100 penetration grade asphalt cement from IFH Bahir Dar, and crushed electric arc furnace slag. Marshall specimens were prepared with bitumen contents ranging from 3.5 % to 5.5 % (at 0.5 % intervals), with slag filler replacing stone dust at 2.5–10 %. Prior to mixing, all materials underwent quality tests such as gradation, specific gravity, abrasion, ductility, penetration, and softening point. Aggregates were blended to meet the target gradation, and mixing was conducted at 145°C, followed by compaction using 75 blows per side. After cooling, specimens were tested for Bulk Density, voids, stability, and flow in accordance with the Marshall method, as shown in Fig. 3.

2.2. Random Forest (RF)

Random Forest (RF), introduced by Breiman (2001), is a robust ensemble learning method that constructs multiple decision trees to enhance prediction accuracy and facilitate variable selection [77]. By employing both bootstrap sampling (row subsampling) and random feature selection (column subsampling), RF effectively mitigates overfitting and handles high-dimensional data [78]. In regression applications, each tree partitions the feature space into regions, and predictions are obtained by averaging the outputs of the ensemble, with the mean squared error (MSE) guiding optimal split decisions [79]. Although individual trees exhibit similar bias, the aggregation process significantly reduces variance, resulting in a more reliable model.

Random Forest (RF) has gained significant recognition for its robustness in handling missing data and outliers, coupled with its computational efficiency. These strengths have established RF as a reliable and versatile non-parametric ML method across a wide range of disciplines, including construction management and transportation engineering. In the transportation sector specifically, RF has been successfully utilized in numerous applications. For example, Kebede et al. [80] employed RF to predict real-time pavement

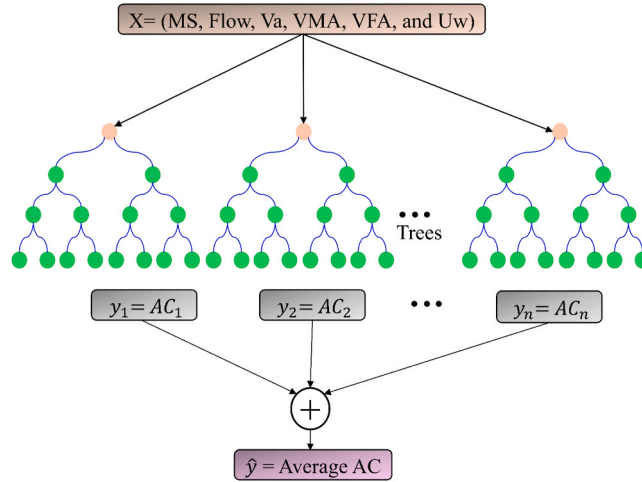


Fig. 4. Illustration of the random forest regression algorithm for asphalt content prediction.

temperature, while Guo et al. [81] used it to assess asphalt performance and evaluate the International Roughness Index (IRI) of asphalt pavements. Gong et al. [82] further optimized the RF algorithm to enhance predictions of the pavement surface condition index, building upon earlier work that applied RF to forecast long-term pavement performance. Collectively, these studies underscore the practical utility and predictive strength of the RF model in infrastructure-related problems. RF's ability to manage high-dimensional datasets with limited sample sizes makes it especially valuable in complex, data-sparse, or privacy-sensitive contexts. Additionally, its inherent capability to handle class imbalances and detect outliers further enhances its suitability for real-world engineering problems where data quality and quantity are often suboptimal [82].

The procedural workflow for constructing a Random Forest model can be delineated as follows:

1. For $f = 1$ to F :

- (a) Draw a random bootstrap sample of size N from the original training set.
- (b) Construct a regression tree T_f by recursively partitioning the dataset according to the following criteria:

- Randomly sample m features from the total set of p features.
- Determine the optimal splitting criterion based on the selected m features
- Partition the node into two sub-nodes. Continue recursive partitioning until a predefined termination criterion (e.g., a minimum node size threshold) is satisfied.

2. Aggregate the predictions from the ensemble of trees $\{T_h\}_i, i = 1, 2, 3, \dots, H$

For a regression task, given a new observation x , obtain predictions from each individual tree and compute the ensemble average as shown in Eq.(1):

$$\hat{y} = \frac{1}{H} \sum_{h=1}^H T_h(x) \quad (1)$$

where H represents the total number of trees in the ensemble. This aggregated prediction serves as the final model output. Fig. 4 provides a conceptual illustration of the Random Forest regression algorithm and its procedural execution.

2.3. DNN

Deep Neural Networks (DNNs), an advanced subset of Artificial Neural Networks (ANNs), leverage deep architectures with multiple interconnected layers to hierarchically process and extract complex features from data [72]. Inspired by human neural mechanisms, ANNs enable self-learning without explicit programming, with DNNs extending these capabilities through additional depth, enhancing their ability to model intricate data structures. A standard DNN comprises an input layer for data preprocessing, hidden layers for hierarchical feature extraction, and an output layer for predictions, where neurons apply weighted transformations, biases, and activation functions to introduce nonlinearity and capture sophisticated patterns [83], as illustrated in Fig. 5. Training via back-propagation iteratively refines weights to minimize predictive error, improving generalization to unseen data. Excelling in image classification, predictive modeling, and time-series forecasting, DNNs handle high-dimensional data synthesis and multimodal

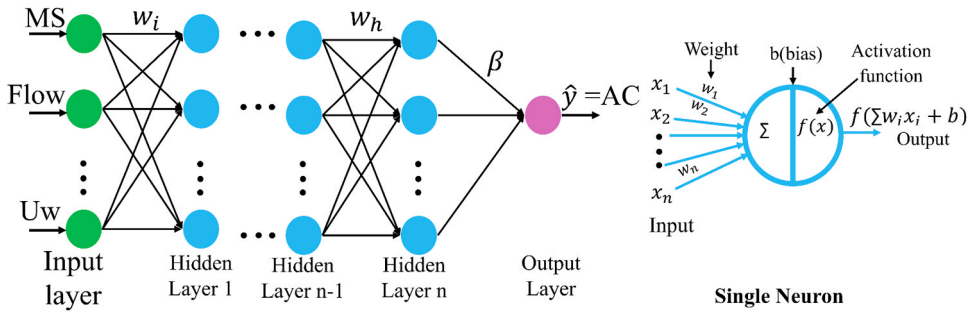


Fig. 5. The architectural layout of DNN models.

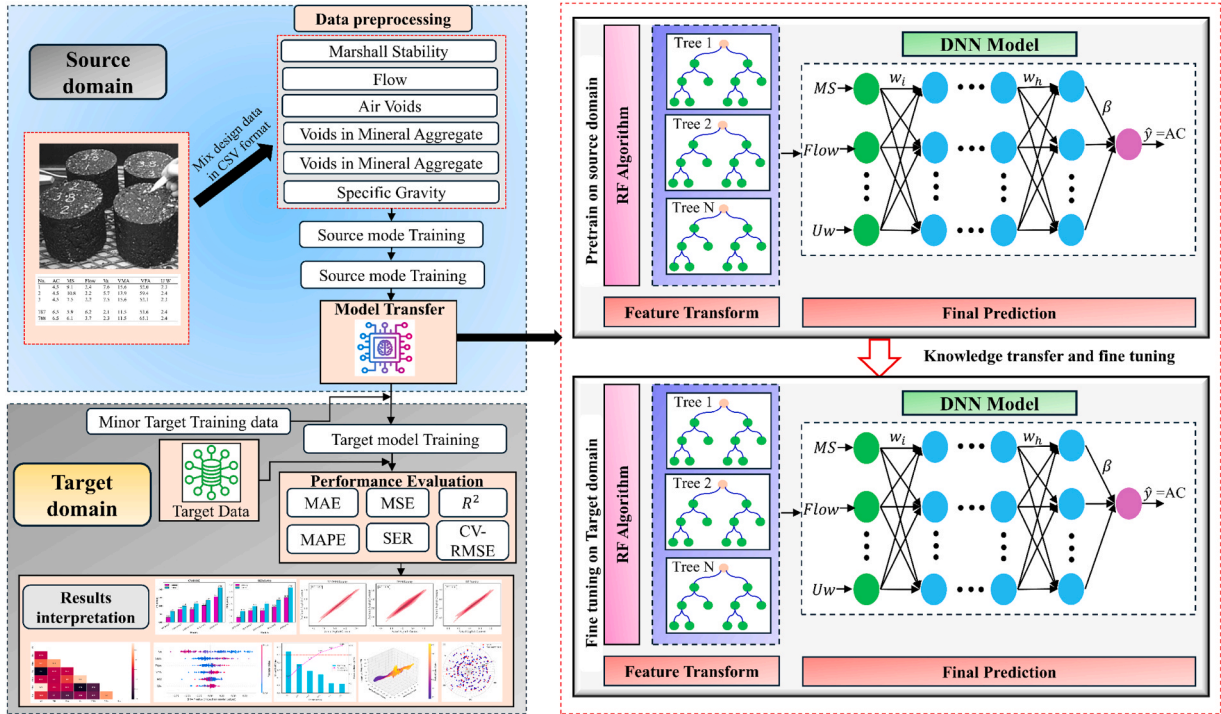


Fig. 6. Research framework of asphalt content prediction.

integration [64,84]. Applications include the prediction of construction cable forces in CFST arch bridges [85], railway track geometry forecasting [86], bridge damage assessment [87] and concrete fatigue life prediction [88]. To enhance performance and efficiency, hyperparameter tuning techniques, such as Random Search, optimize layers, neuron allocations, activation functions, and learning rates, ensuring robust and scalable fully connected architectures.

Further, advanced deep learning frameworks provide sophisticated debugging tools and facilitate modular model extension, thereby enhancing overall development efficiency [89]. For a DNN with hidden layers, the output is defined by the functional mapping presented in Eq. (2).

$$x_i^1(k) = f\left(\sum_{l=1}^{nl} w_{ij}^1 x_i^{l-1}(k) + a_i^1\right) \quad (2)$$

Here, $k = 1, 2, \dots, N$ indexes the input samples; $l = 1, 2, \dots, h+1$ indexes the layers (with h hidden layers); k and j are the neuron indices in the current and preceding layers, respectively; w is the weight matrix; a is the bias; and f is the activation function [76], as formulated in Eq.(3).

$$f(x) = \max(0, x) \quad (3)$$

The output layer's computation is expressed in Eq. (4).

$$\hat{y}(k) = \sum_{j=1}^{nh} \beta_j f \left(\sum_{i=1}^{nh} w_{ij}^h x_i^h(k) + a_j^h \right) + b \quad (4)$$

Here, $\hat{y}$ denotes the DNN output, f is the activation function, β is the output layer's weight matrix, and ω is its bias term.

2.4. Hybrid RF-DNN transfer learning

This study proposes a hybrid (RF-DNN) model designed for transfer learning in regression tasks. The selection of this specific architecture is theoretically grounded and practically motivated by the problem's core challenges and the nature of the dataset. RF, as the input data consists of structured, tabular measurements from laboratory experiments, for which RF is exceptionally well-suited. RF excels at capturing complex, non-linear interactions and is inherently robust to the noise and outliers common in experimental settings, serving as a powerful front-end for feature extraction. The model's back-end employs a DNN which is essential for the transfer learning strategy. This approach requires a neural network to implement the gradient reversal mechanism needed to learn domain-invariant representations and align feature distributions between the source and target domains. This makes the DNN a crucial component for enabling knowledge transfer. Alternative architectures like Convolutional Neural Networks (CNNs) or Long Short-Term Memory (LSTM) networks were deemed unsuitable, as they are designed for data with spatial or sequential structures, which are absent in our feature set.

The procedural workflow of the RF-DNN model (Fig. 6) begins with data collection and preprocessing, which involves imputation, normalization, and one-hot encoding. Next, the RF component is trained on the source dataset to extract key nonlinear features and generate a set of informative, transformed features. These are then concatenated with the original source features to train the DNN regression predictor. Simultaneously, the DNN is trained adversarially to align the source and target feature distributions, ensuring domain-invariant representations. To transfer the model to the target domain, the pre-trained RF's feature transformation is applied to the target data, creating a consistent feature space. These transformed target features, combined with the original target features, are then fed into the already-trained DNN predictor. Finally, the model's performance on the target domain is evaluated using standard regression metrics. This synergistic approach effectively combines RF's robust feature selection with the DNN's advanced cross-domain adaptation capabilities, ensuring robust and transferable predictions across varying asphalt mix compositions and testing environments.

2.5. Model performance evaluation

Model evaluation is fundamental to predictive analytics, addressing key questions: (i) Does the model fit the data? (ii) Which model performs best? and (iii) How similar are the models? In this study, Mean Squared Error (MSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Coefficient of Variation of Root Mean Squared Error (CV-RMSE) are utilized to assess model performance. And also, Goodness-of-fit statistics, including the coefficient of determination (R^2) and Standard Error Ratio (S_e/S_y), assess model performance and are computed for each predictive model using the following equations.

These evaluation metrics are computed using Eqs. (5)-(11), respectively.

$$MAE = \sum_{n=1}^N \frac{|y_i - \hat{y}_i|}{N} \quad (5)$$

$$MAPE = \sum_{n=1}^N \frac{|y_i - \hat{y}_i|^2}{y_i} \quad (6)$$

$$MSE = \sum_{n=1}^N \frac{|y_i - \hat{y}_i|^2}{N} \quad (7)$$

$$RMSE = \sqrt{\sum_{n=1}^N \frac{|y_i - \hat{y}_i|}{N}} \quad (8)$$

$$CV \text{ RMSE} = \frac{RMSE}{\bar{y}} \quad (9)$$

$$R^2 = 1 - \frac{\sum_{n=1}^N (y_i - \hat{y}_i)^2}{\sum_{n=1}^N (y_i - \bar{y})^2} \quad (10)$$

$$\frac{S_e}{S_y} = \frac{RMSE}{\sigma_y} \quad (11)$$

where y_i denotes the actual site observed AC, $\bar{y}_i$ indicates the average value and $\hat{y}_i$ represents the predicted AC and $\bar{y}$ the mean of the

Table 1
Statistical summary of experimental dataset.

Variables	Unit	Mean	Std	Min	25 %	50 %	75 %	Max
AC	%	5.41	0.80	3.00	5.00	5.50	6.00	7.00
MS	kN	9.29	4.38	1.00	6.10	8.78	11.30	34.96
Flow	mm	4.92	3.04	1.10	3.00	3.85	5.10	20.00
Va	%	3.85	2.70	0.18	1.70	3.70	5.30	14.55
VMA	%	15.68	4.52	10.10	13.20	14.52	16.14	36.01
VFA	%	72.79	12.99	32.43	63.11	71.90	82.83	98.90
Density (γ)	g/cm ³	2.36	0.12	1.59	2.31	2.39	2.43	2.58

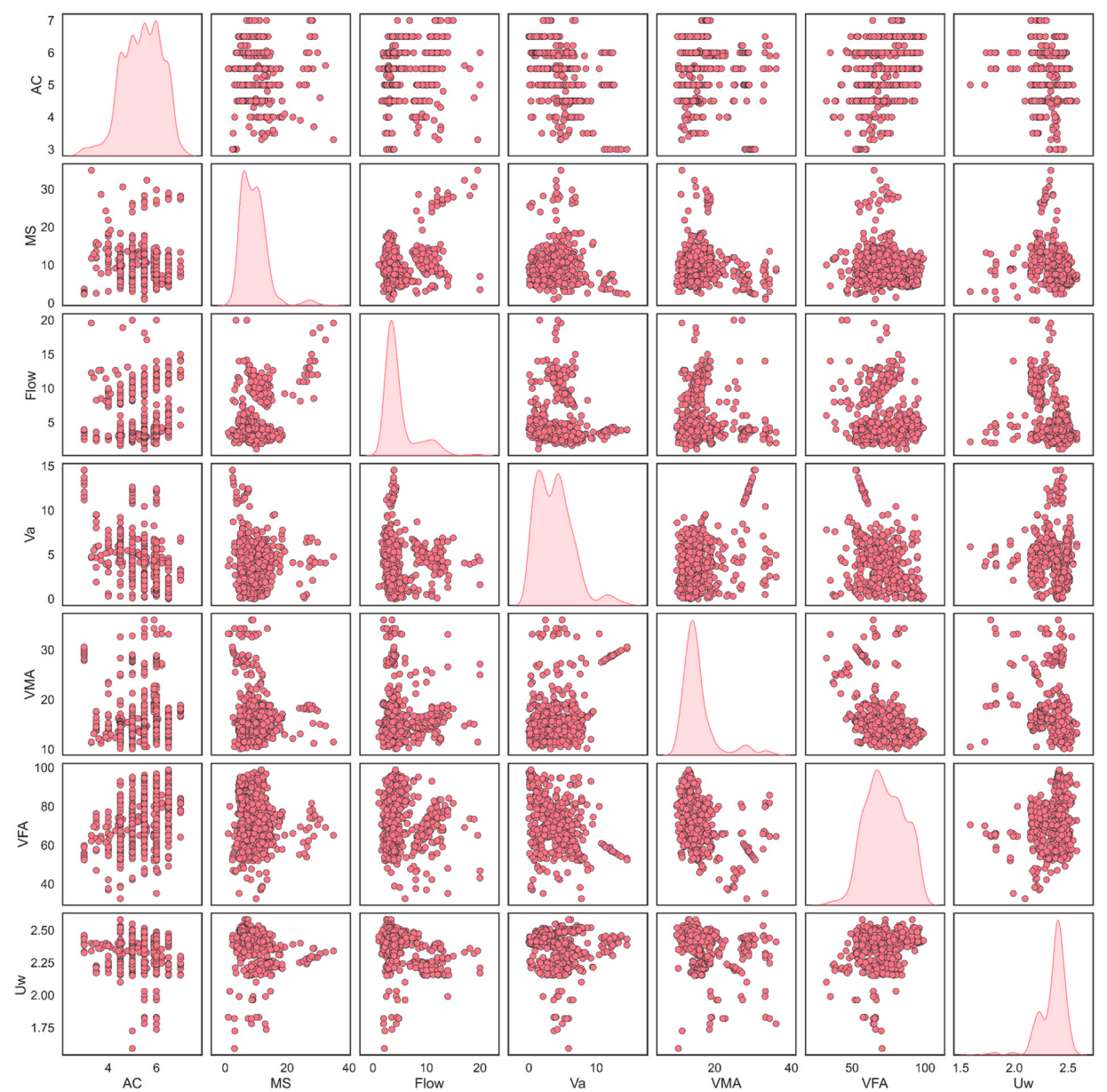


Fig. 7. Pairplot Visualization of Asphalt Mix Properties Showing Relationships Between Asphalt Content, Stability, Flow, Air Voids, VMA, VFA, and mix density for source dataset.

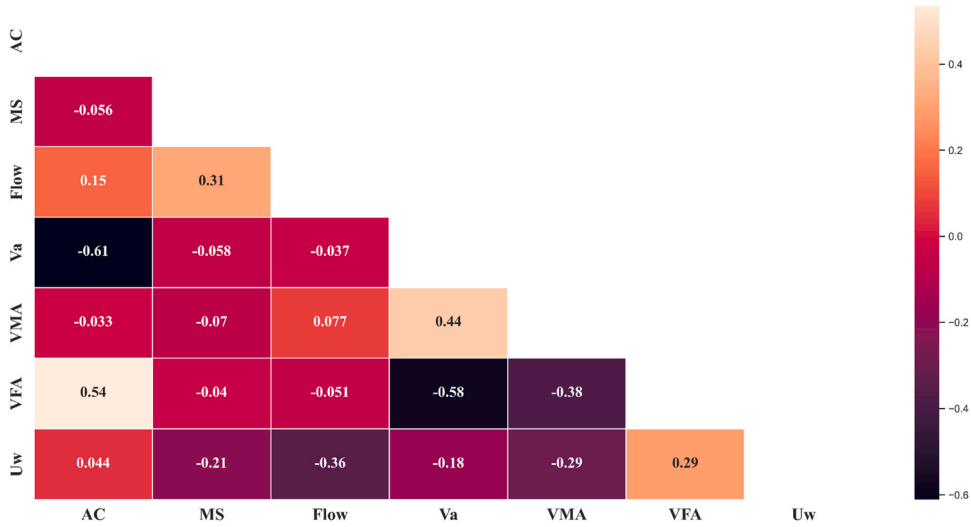


Fig. 8. Displays the Pearson correlation coefficient matrix for the source datasets.

observed AC values, N is the total number of observations and σ_y = Standard deviation of observed values.

The MSE quantifies the average squared deviation between predicted and actual AC, while MAE captures the absolute average discrepancy in AC predictions. Mean MAPE evaluates the relative predictive error in percentage terms, and the Coefficient of CV-RMSE normalizes RMSE against the mean observed AC, facilitating cross-comparisons. R^2 assesses the strength of correlation between predicted and measured AC values, whereas Se/Sy provides a direct measure of model uncertainty, reflecting predictive reliability in asphalt content estimation. Lower MSE, MAE, MAPE, and CV-RMSE indicate superior predictive performance, while higher R^2 and lower Se/Sy enhance model accuracy. Pellinen [90] established standardized evaluation criteria, widely adopted in predictive research and applied in this study: models are rated Excellent with $R^2 \geq 0.90$ and Se/Sy ≤ 0.35 ; Good for R^2 between 0.70 and 0.89 and Se/Sy between 0.36 and 0.55; Fair for R^2 between 0.40 and 0.69 and Se/Sy between 0.56 and 0.75; and Poor or Very Poor for lower R^2 values.

2.6. Shapley Additive Explanations (SHAP) for Model Interpretability

MLmodels are often considered "black boxes" due to their complexity, making interpretation challenging. SHAP (Shapley Additive Explanations), introduced by Lundberg and Lee [91], is a model-agnostic method that quantifies feature contributions to predictions using cooperative game theory [92]. For a model with input features $x_1, x_2, \dots, x_m$ the interpretation function is:

$$g(x') = \phi_o + \sum_{i=1}^M \phi_i x'_i \quad (12)$$

Where M is the number of features, ϕ_o is a constant, and ϕ_i represents each feature's influence. SHAP provides clear attributions, enhancing model transparency and interpretability [83]).

3. Results and discussions

The proposed method was validated using a Marshall Mix Design dataset compiled from extensive literature sources, aimed at evaluating model performance. This dataset includes six independent variables MS, Flow, Va, VMA, VFA, and Uw with AC serving as the dependent variable. The AC values averaged 5.41 % with a standard deviation of 0.80 %, ranging from 3.00 % to 7.00 %, indicating diverse mix proportions. MS demonstrated considerable variability (mean 9.29 kN, maximum 34.96 kN), reflecting material property and compaction differences, while Flow ranged from 1.10 to 20.00 mm (mean 4.92 mm), indicating variability in deformation behavior. Va (mean 3.85 %, range 0.18–14.55 %) and VMA (10.10–36.01 %) offered insights into mix durability and compaction characteristics. VFA, with an average of 72.79 % (range 32.43–98.90 %), suggested significant variation in binder absorption and aggregate interlocking. Lastly, the Uw ranged from 1.59 to 2.58 g/cm³ (mean 2.36 g/cm³), capturing variability in material composition and compaction level. A detailed distribution of the dataset sources is provided in Table 1.

Figs. 7 and 8 illustrate key relationships in Marshall Mix Design, highlighting trade-offs in optimizing asphalt content (AC). The pairplot (Fig. 7) provides a detailed visualization of the relationships between key variables in asphalt mix design, including Asphalt Content (AC), Marshall Stability (MS), Flow, Air Voids (Va), Voids in Mineral Aggregate (VMA), Voids Filled with Asphalt (VFA), and Bulk Density (Uw). The univariate distributions along the diagonal reveal skewed or multimodal patterns, indicating potential variability. Strong positive correlations between VFA and VMA, as well as Uw and VMA, suggest that higher VMA enhances asphalt filling and increases mixture density, while negative correlations between Va and VFA, and MS and Va highlight that reducing air voids

Table 2
Optimized Hyperparameters of the DNN Model Using Bayesian Optimization.

Model	Hyperparameter	Optimized
DNN	Number of neurons in hidden layers	(256, 16, 54)
	Activation function	ReLU
	Optimizer	Adam (lr = 0.00415)
	Epochs	120
	Batch size	32
	Loss function	MSE
	Metrics	RMSE
RF	n_estimators	300
	max_depth	30
	max_features	2
	min_samples_leaf	1
	min_samples_split	2
	random_state	42

Table 3
Predictive performance different ML and DL models.

Model	Training Set Evaluation						Testing set evaluation					
	MAE	MSE	R2	MAPE	CV-RMSE	SER	MAE	MSE	R2	MAPE	CV-RMSE	SER
RF	0.08	0.01	0.98	1.58	2.25	0.15	0.12	0.02	0.96	2.24	2.81	0.21
LGM	0.17	0.06	0.91	3.38	4.43	0.29	0.15	0.03	0.94	2.74	3.36	0.25
DNN	0.04	0.00	0.99	0.79	1.12	0.08	0.14	0.03	0.95	2.52	3.35	0.21
RF-DNN	0.05	0.01	0.99	1.03	1.46	0.10	0.09	0.02	0.97	1.67	2.30	0.17
XGB	0.20	0.10	0.85	4.01	5.80	0.38	0.20	0.10	0.85	4.01	5.80	0.38
AdaBoost	0.35	0.17	0.74	6.67	7.66	0.51	0.30	0.11	0.79	5.52	6.16	0.46
SVR	0.22	0.10	0.85	4.30	5.96	0.39	0.26	0.14	0.73	4.84	6.94	0.52
ANN	0.21	0.08	0.88	4.11	5.28	0.35	0.27	0.16	0.70	5.07	7.32	0.55
LSTM	0.29	0.16	0.76	5.79	7.38	0.49	0.33	0.21	0.61	6.09	8.36	0.63
Linear	0.40	0.29	0.57	8.00	9.97	0.66	0.40	0.29	0.57	8.00	9.97	0.66

improves stability and durability. A mild positive correlation between Flow and AC suggests that increased asphalt content slightly raises deformation potential. Cluster patterns in Va vs VMA and Flow vs MS indicate possible subgroups, likely reflecting different mix gradations or compaction efforts, with potential outliers pointing to inconsistencies in material properties or compaction techniques. These insights emphasize the need to balance AC, Va, VMA, and VFA to optimize mix performance, minimize deformation, and enhance durability. Leveraging ML models, especially transfer learning-based approaches, can further improve predictive accuracy and optimize mix designs in resource-constrained environments. This analysis offers valuable guidance for refining asphalt mix proportions to ensure better pavement performance under real-world conditions.

The correlation matrix (Fig. 8) further reveals a weak link between AC and Marshall Stability (MS: -0.056), indicating that aggregate gradation and void structure play a more dominant role in stability than binder content alone. Flow increases with AC (0.15), affecting deformation resistance, while Va's inverse correlation with MS (-0.058) underscores the importance of balanced void ratios for mix stability. Additionally, AC has a negligible impact on VMA (-0.033), showing that void space in the aggregate structure is largely governed by compaction and gradation rather than binder content. The strong negative correlation between VMA and VFA (-0.38) suggests that excessive binder filling voids may lead to bleeding and skid resistance loss. With density weakly correlated with AC (-0.29), it is evident that density regulation is more influenced by aggregate properties than binder content. Given these complex interdependencies, ML models and SHAP analysis offer a data-driven approach to refine binder selection, enhancing mix durability, cost-effectiveness, and climate resilience as discussed in Section 3.2.

Following data preprocessing, initial baseline models, RF and DNN, were developed. To enhance model performance, advanced Bayesian Optimization (BO) was employed for hyperparameter tuning. First, BO with cross-validation was applied to the RF model, optimizing critical hyperparameters related to feature selection and transformation. Subsequently, the DNN model underwent BO-based tuning, refining layer depth, neuron allocation, activation functions (ReLU), and regularization strategies (dropout, batch normalization). Early stopping was incorporated to mitigate overfitting. The optimized hyperparameters for both RF and DNN are presented in Table 2. Experiments were conducted on an HP Z6 G5 workstation (Intel® Xeon® W5-3425, 192 GB RAM), utilizing Python 3.12.7, PyTorch 2.6.0 +CPU, and Anaconda (Conda 24.11.3). A comprehensive multi-criteria evaluation was performed to assess the proposed RF-DNN hybrid model and established regression-based predictive models. Evaluation metrics included Mean MAE, MSE, R^2 , MAPE, CV-RMSE, SER) and SHAP values. For model testing, 20 % of the source dataset was used. In the fine-tuning transfer phase, 60 % of the target dataset was utilized.

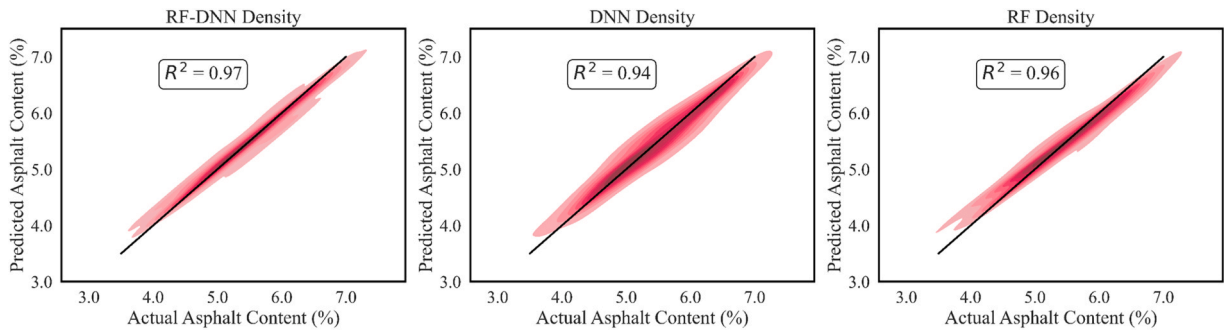


Fig. 9. Comparison of RF-DNN, DNN, and RF model performance in asphalt content prediction.

3.1. The pre-trained RF-DNN

In this phase, the RF-DNN hybrid model is trained and validated on the source dataset to assess its predictive capability for asphalt content (AC) and enable transfer learning. The RF component performs effective feature selection by capturing nonlinear relationships, while the DNN extracts high-level latent patterns for accurate prediction. This step is essential for evaluating the model's performance and its ability to generalize across key variables such as VFA, Flow, Va, VMA, MS, and Uw. Table 3 summarizes the performance of various ML and DL models, highlighting the superiority of the proposed RF-DNN architecture.

The RF-DNN hybrid model stands out as the optimal choice due to its exceptional balance of accuracy and generalization, evidenced by a training R^2 of 0.99 and a testing R^2 of 0.97, along with minimal errors (training MAE of 0.05, testing MAE of 0.09, MAPE of 1.67, and CV-RMSE of 2.30). While DNN closely follows with similar accuracy, RF-DNN demonstrates superior resistance to overfitting. RF and LGM offer reliable performance but exhibit slightly higher errors. XGB, SVR, and ANN provide moderate results, with increased error rates and weaker generalization, particularly evident in ANN's significant drop in testing R^2 . AdaBoost, LSTM, and Linear Regression are the weakest performers, displaying poor generalization and high error rates, making them unsuitable for accurate predictions. Consequently, RF-DNN emerges as the preferred model for robust and reliable predictions, excelling in both accuracy and generalization compared to the other evaluated models.

Fig. 9 presents a comparative analysis of three models: the hybrid RF-DNN and the baseline models, DNN and RF, in predicting asphalt content. The density plots illustrate the relationship between actual and predicted values, with the red dashed line representing perfect predictions ($y = x$ line). Among the three models, RF-DNN demonstrates the highest predictive accuracy, achieving an R^2 value of 0.97, indicating that 97 % of the variance in actual asphalt content is explained by the model. The density contours for RF-DNN are tightly clustered around the ideal prediction line, signifying high precision and minimal error. Comparatively, the DNN model ($R^2 = 0.94$) exhibits slightly greater deviation from the perfect prediction line, suggesting a marginally lower predictive performance. The RF model alone achieves an R^2 of 0.95, surpassing DNN but still trailing behind the RF-DNN hybrid. This finding suggests that ensemble-based approaches, particularly hybrid models like RF-DNN, enhance predictive accuracy by leveraging both decision tree-based learning and deep learning capabilities. Overall, the RF-DNN model outperforms both standalone RF and DNN models, making it the most reliable choice for asphalt content prediction due to its superior generalization ability and reduced prediction variance.

Fig. 10 compares the performance of four models in predicting asphalt content: RF, DNN, an RF-DNN hybrid model, and Linear Regression. The red squares represent measured values, while the blue dots denote model-predicted values. Among these models, the RF-DNN hybrid model demonstrates the highest consistency with the measured values, effectively integrating RF's feature selection capabilities with DNN's nonlinear fitting strengths, resulting in superior generalization and predictive performance. The model accurately captures the underlying data patterns, although minor deviations are observed, particularly at the outer edges where extreme asphalt content values (3.5 % and 5.5 %) occur. These deviations may be attributed to nonlinear interactions between input variables, such as MS, Flow, Va, VFA, VMA, and Uw, or potential inconsistencies and noise in the laboratory data. Residuals, representing the differences between actual and predicted values, are randomly distributed, indicating no systematic bias. However, larger discrepancies are noted at higher AC values (near 6.5 %) and lower AC values (near 3.5 %), suggesting increased difficulty in capturing extreme data points. Despite these deviations, the predicted values remain tightly clustered around the actual values, particularly within the 4.0–5.5 % range, where the model demonstrates robust performance. Comparatively, DNN performed second best, RF exhibited slightly higher prediction errors, and the linear model performed the worst due to its inability to capture the nonlinear relationships inherent in the data. Overall, the RF-DNN model exhibits exceptional predictive accuracy and strong potential for application in complex asphalt content prediction tasks.

Fig. 11 further illustrates the predictive performance of the RF-DNN hybrid model, demonstrating a strong overall correlation between actual and predicted asphalt content, with a diagonal trend indicating high accuracy and generalization. The color gradient reflects the predicted values, where smoother transitions in the lower asphalt content range (4.0–5.0) suggest stable predictions, while fluctuations and spikes in the mid-to-high asphalt content range (5.5–6.5) indicate localized prediction variability, potentially caused by data noise, outliers, or model sensitivity. The model exhibits optimal predictive performance within the intermediate asphalt content range (4.5–5.5 %), where minimal deviations from actual values are observed. However, the higher variability at the boundaries suggests the increased difficulty in accurately predicting extreme asphalt content values, which may be attributed to the

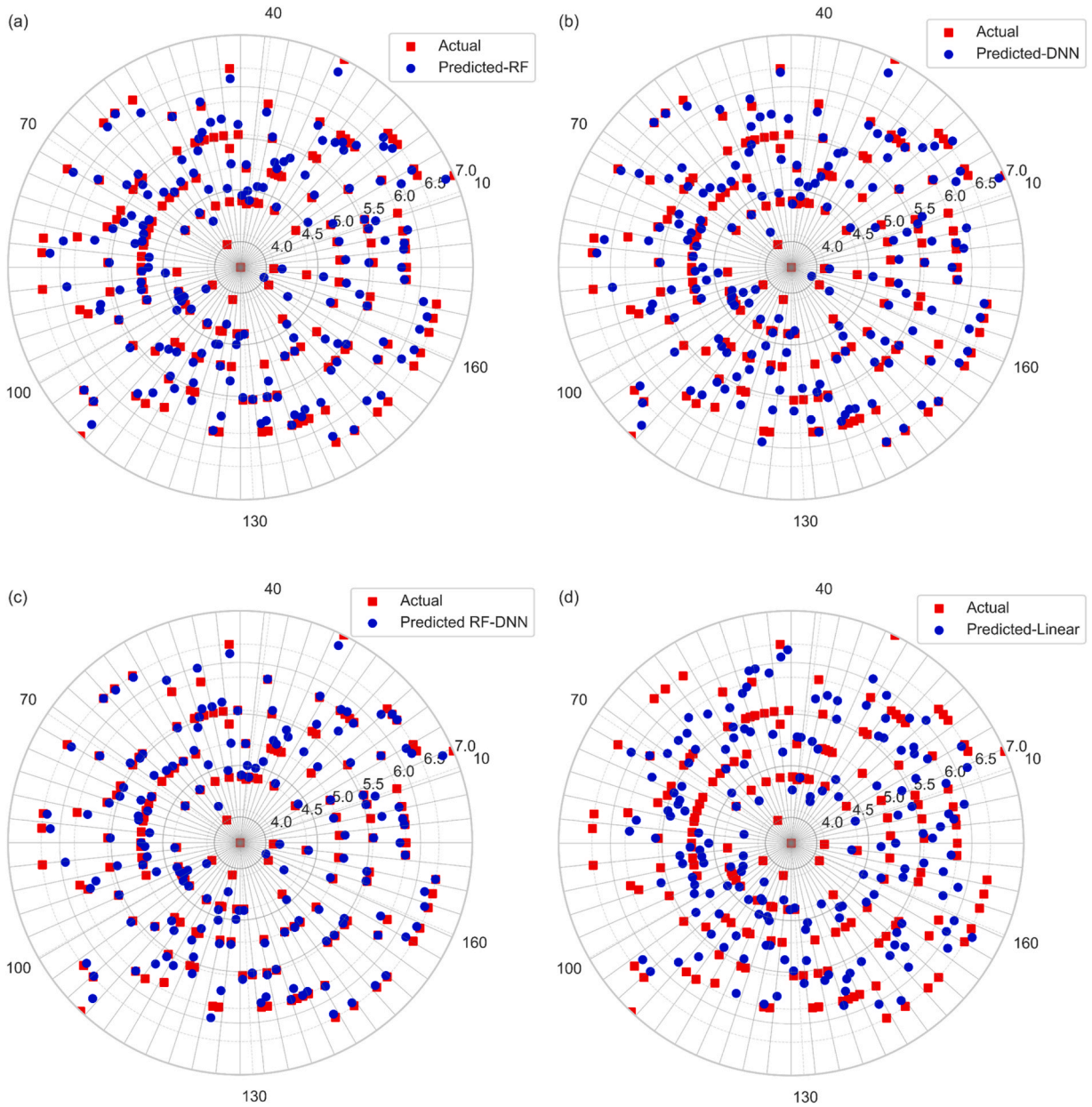


Fig. 10. Model Performance in Asphalt Content Prediction: Actual vs. Predicted values.

sensitivity of the mix design parameters in these ranges. The random distribution of errors suggests the absence of systematic bias, underscoring the reliability and robustness of the model in capturing the intricate relationships between input variables and asphalt content. These findings highlight the model's ability to generalize well across diverse datasets while maintaining high predictive accuracy, making it a valuable tool for optimizing asphalt mix designs.

3.2. Global interpretation

Global interpretation provides an overall view of how input features influence the model's predictions. The SHAP summary plot in Fig. 12 reveals the most influential features, highlighting Va, VMA, and Flow as primary drivers, while Uw and MS contribute moderately. A gradient from blue to red represents low to high feature values, respectively. The plot shows that Va and Flow are inversely correlated with predictions, suggesting that higher values reduce the output, whereas VMA exhibits a positive correlation. The more complex distributions observed for VFA and MS imply potential non-linear interactions. Scientifically, this underscores the critical role of void structures in asphalt mix performance, with the inverse Va–Uw relationship suggesting a density trade-off. The non-

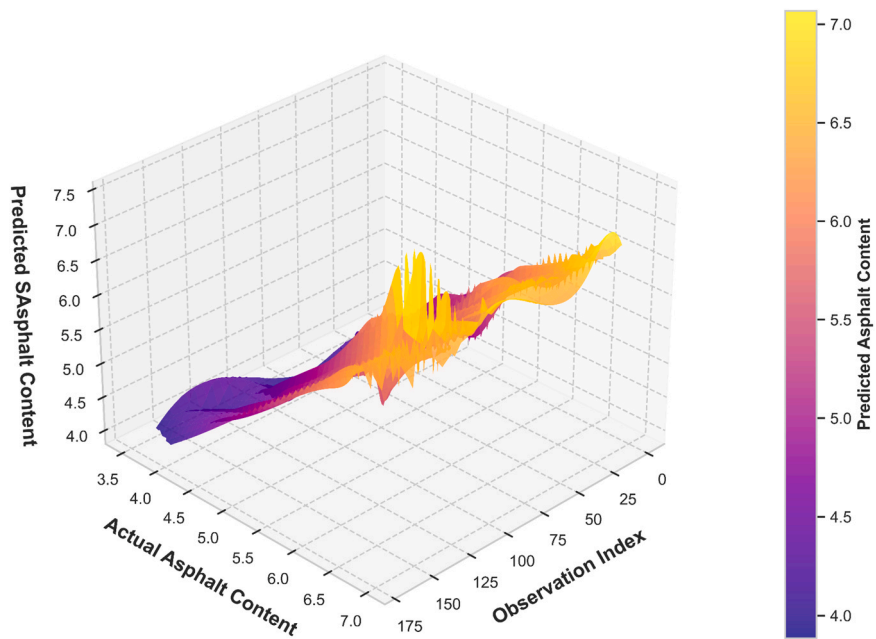


Fig. 11. 3D Surface Plot of Predicted vs. Actual Asphalt Content.

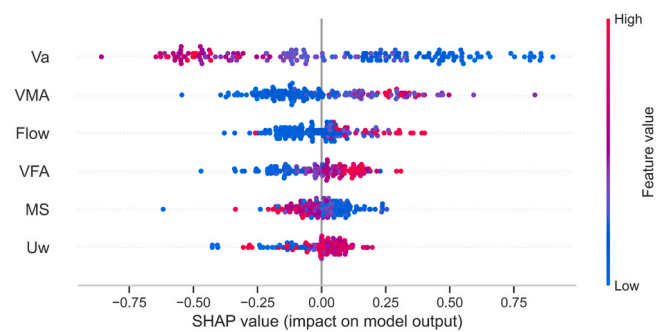


Fig. 12. SHAP Summary Plot of Feature Importance.

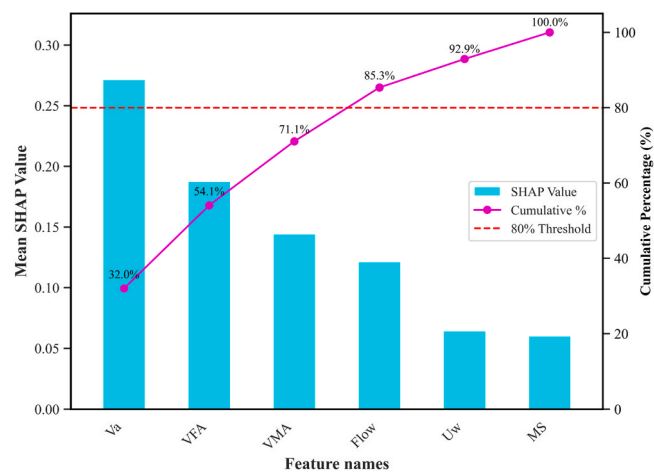


Fig. 13. Global feature importance for the feature parameters.

Table 4
Comparative Analysis of Model Accuracy for Asphalt Content prediction.

Model	Training Set Evaluation						Testing set evaluation					
	MAE	MSE	R2	MAPE	CV RMSE	SER	MAE	MSE	R2	MAPE	CV RMSE	SER
RF-DNN-T	0.03	0.01	0.98	0.64	1.21	0.08	0.08	0.02	0.97	1.61	2.68	0.18
XGB-DNN-T	0.09	0.02	0.95	2.16	3.20	0.19	0.12	0.03	0.93	2.66	4.01	0.26
LGBM-DNN-T	0.11	0.02	0.96	2.42	3.16	0.19	0.15	0.04	0.91	3.29	4.56	0.30
SVR-DNN-T	0.12	0.03	0.94	2.57	4.12	0.25	0.19	0.06	0.87	4.00	5.37	0.36
Adda-DNN-T	0.22	0.08	0.84	5.08	6.23	0.40	0.31	0.16	0.68	6.81	8.46	0.56

linear effects of Flow and VFA, as reflected in the SHAP values, indicate the need for advanced modeling approaches to optimize mix design and support data-driven performance improvements.

Fig. 13 presents a cumulative SHAP analysis for AC prediction, highlighting the relative importance of six key features. Va dominates the prediction, contributing 32 % by influencing mix stability and durability. VFA follows with 22 %, emphasizing its role in enhancing mix durability and preventing deformation issues, such as rutting or bleeding. VMA contributes 17 %, ensuring adequate void space for bitumen and air, thereby impacting long-term performance. Flow accounts for 14 %, reflecting its influence on deformation resistance and workability. Uw and MS contribute 9 % and 6 %, respectively, indicating their comparatively smaller role in predicting asphalt content despite their impact on mechanical performance.

The cumulative analysis shows that Va, VFA, and VMA collectively contribute 71.1 % of the predictive power, with Flow increasing this to 85.3 %, surpassing the critical 80 % threshold. This suggests that while incorporating all features improves prediction accuracy, the additional features (Uw and MS) yield diminishing returns. The dominance of Va and VFA underscores the importance of managing void structures to enhance mix durability and minimize performance variability, while the lower contributions of MS and Uw confirm their secondary role in determining AC. This SHAP-based feature ranking provides actionable insights for optimizing asphalt mix designs by focusing on the most influential parameters.

3.3. The Fine-tuned RF-DNN for AC prediction

In the next stage, the second key contribution of this study transfer learning for laboratory effort reduction is investigated. To simulate a realistic low-resource scenario, it is assumed that only 40 % of the target dataset, obtained through laboratory experiments, is available for training, while the remaining 60 % is reserved exclusively for testing. This setup enables a robust evaluation of the model's ability to transfer knowledge from the source domain and generalize effectively to the target domain, thereby reducing the need for extensive experimental data collection. To assess transfer performance in AC prediction, a hybrid RF-DNN model is developed. The model's hyperparameters are optimized, including the number of fully connected layers, neurons per layer, learning rate, activation functions, and optimization algorithm. Additionally, the RF feature extractor is fine-tuned by adjusting the number of trees, tree depth, and feature selection strategy to enhance domain-invariant feature representation. Table 4 presents a comparative analysis of model accuracy for AC prediction, offering a quantitative assessment of transfer learning performance.

Table 4 presents the evaluation of five hybrid transfer learning models for asphalt content prediction, assessed using multiple statistical metrics across training and testing sets. Among them, the RF-DNN-T model demonstrates the most robust performance, achieving the lowest testing errors (MAE of 0.08, MSE of 0.02, and MAPE of 1.61) and the highest R^2 of 0.97, indicating excellent accuracy and generalization. Ensemble-based hybrids like XGB-DNN-T and LGBM-DNN-T also perform well, with testing R^2 values of 0.93 and 0.91, respectively, though they show slightly higher error rates compared to RF-DNN-T. SVR-DNN-T exhibits moderate performance, with R^2 of 0.87 and increased error metrics. The Adda-DNN-T model demonstrates the weakest predictive reliability, reflected in the highest error rates (MAE of 0.31 and MAPE of 6.81) and the lowest R^2 of 0.68 on the testing set. These results underscore the effectiveness of hybrid ensemble-DNN models particularly RF-DNN-T in capturing the nonlinear relationships inherent in asphalt content prediction, establishing them as the most accurate and reliable options among the evaluated approaches.

3.4. The transfer performance comparison

In this section, RF is employed as the feature extractor within the proposed RF-DNN hybrid transfer framework. Additionally, alternative ML models including XGB, LGBM, SVR, and AdaBoost are explored as comparative feature extractors. To assess the impact of different domain adaptation network architectures, a fully connected DNN is also implemented as a baseline feature extractor, while the regression predictor and domain classifier components remain consistent across all models. As a result, various hybrid DNN-based architectures XGB-DNN, LGBM-DNN, AdaBoost-DNN, and SVR-DNN are developed and benchmarked against the proposed RF-DNN model for asphalt content prediction. Among all hybrid models, the RF-DNN transfer model demonstrates the highest predictive accuracy. Its ability to effectively capture complex feature interactions and perform robust feature selection enables it to outperform all other models. The results further validate RF-DNN as the most effective domain adaptation model, attributed to its superior capacity to extract meaningful and transferable representations essential for AC prediction.

Fig. 14 provide a comprehensive performance comparison of five hybrid models RF-DNN, XGB-DNN, LGBM-DNN, SVR-DNN, and Adda-DNN based on six statistical metrics: Mean MAE, MSE, R^2 , MAPE, CV-RMSE, and SER. Among these, the RF-DNN model exhibits consistently superior performance across all metrics. For instance, it achieves the lowest MAE and MSE for both training and testing,

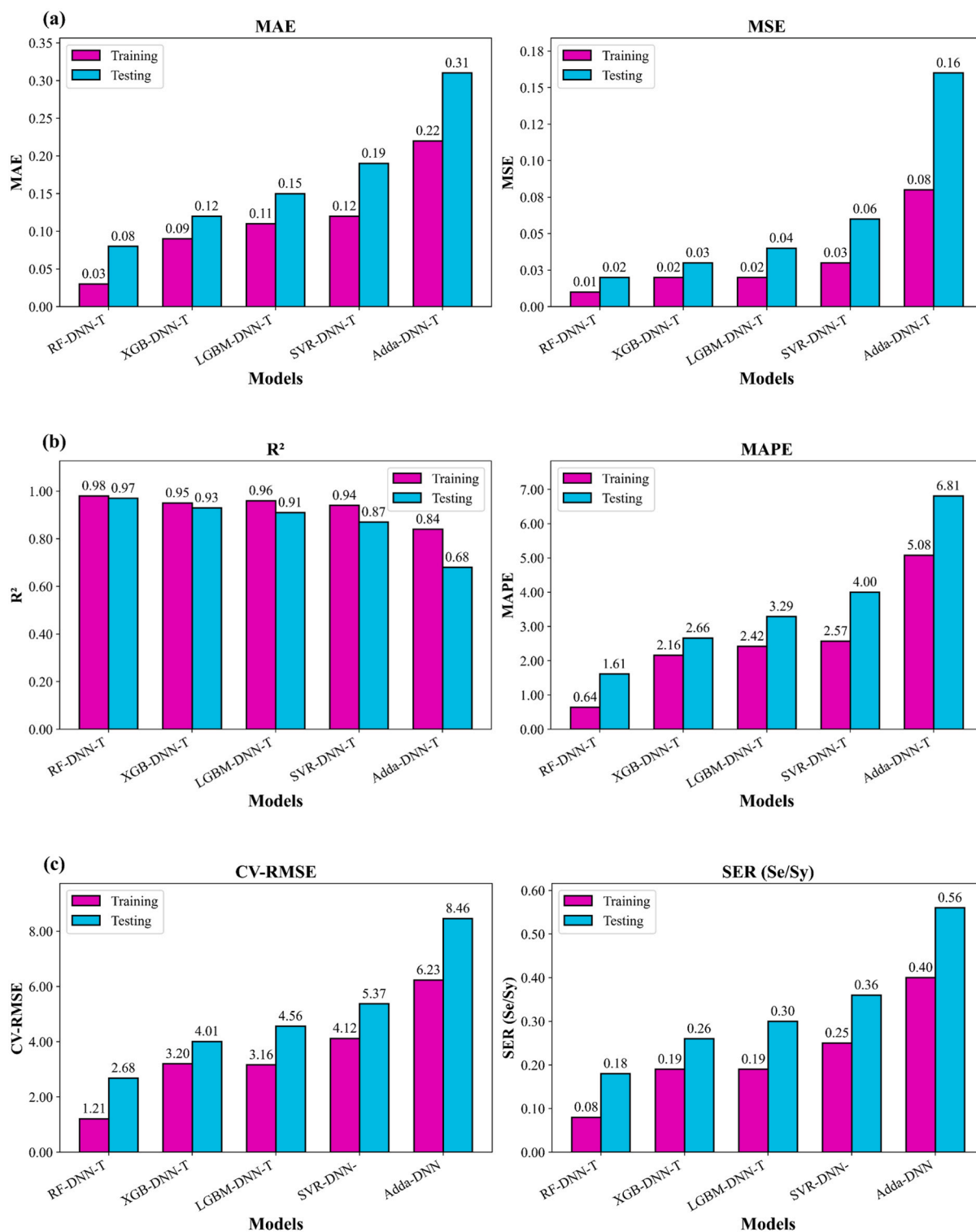


Fig. 14. Comparative evaluation of various hybrid transfer models: (a) MAE and MSE for training and testing datasets; (b) R^2 and MAPE for training and testing datasets; (c) CV-RMSE and SER for training and testing datasets.

indicating that its predictions deviate minimally from actual values on both absolute and squared scales. A high R^2 value demonstrates that the model explains nearly all the variability in the response variable, while the lowest MAPE suggests exceptional predictive accuracy in percentage terms. Furthermore, the smallest CV-RMSE reflects minimal error relative to the mean of the observed values,

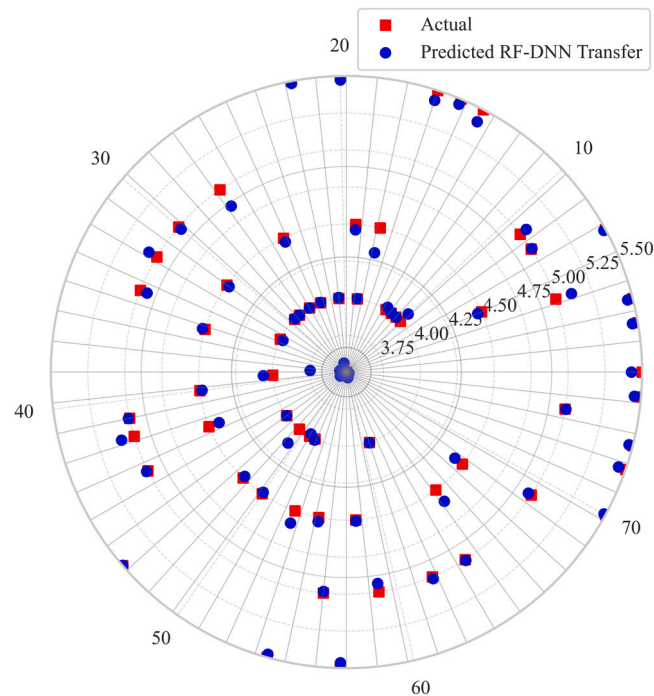


Fig. 15. Polar Scatter Plot: Actual vs. Predicted Values for the RF-DNN Transfer Hybrid Model.

and a small SER implies that the residuals, or errors, have low variability signifying a tight fit of the regression line to the data. In contrast, the Adda-DNN model consistently underperforms with the highest error metrics indicating poor accuracy and generalization. Intermediate results are observed for XGB-DNN, LGBM-DNN, and SVR-DNN, which perform reasonably well but fall short of RF-DNN's robustness. Overall, a small SER, low MAE/MSE/MAPE, and CV-RMSE, combined with a high R^2 , signify a model that not only makes accurate predictions but also generalizes well to unseen data, positioning RF-DNN as the most reliable and scientifically sound hybrid modeling approach among those evaluation.

Fig. 15 illustrates a polar scatter plot comparing actual (red squares) and predicted asphalt content values (blue circles) derived from the RF-DNN Transfer hybrid model. The consistent overlap between the two indicates strong model performance, low prediction error, and minimal systematic bias, demonstrating the model's ability to accurately estimate asphalt content across various samples. The polar scatter plot reveals a strong correlation between actual and predicted values, with minimal errors across most ranges. Points near the center represent lower content values, where the model shows high accuracy. In the moderate content range, dense clustering and close alignment of actual and predicted values further highlight the model's consistent performance. However, at higher content levels (outer region), slight deviations suggest increased error, indicating a potential need for further refinement. Overall, the model demonstrates reliable predictive performance, with only minor variability in higher content regions.

Fig. 16 presents the regression analysis of four hybrid models RF-DNN, XGB-DNN, LGBM-DNN, and SVR-DNN by comparing the predicted asphalt content with the actual asphalt content. The RF-DNN model achieves the highest accuracy with an R^2 of 97 %, indicating excellent predictive capability with minimal deviation, as reflected by the tight clustering of points along the regression line. The XGB-DNN and LGBM-DNN models also perform well, with R^2 values of 93 % and 91 %, respectively, though slight dispersion is observed at higher asphalt content values. The SVR-DNN model shows the lowest performance, with an R^2 of 87 %, and a broader spread of data points, indicating greater prediction error. Overall, RF-DNN demonstrates superior generalization and predictive reliability compared to the other hybrid models.

3.5. Limitations and Future Work

The main limitation of the proposed method is its reliance on a single-source laboratory dataset for fine-tuning, which restricts transferability to regions with different climates, pavement designs, and materials. As a result, the model's generalizability under extreme weather and varied traffic loads remains unverified. The current framework is a proof-of-concept for hybrid transfer learning and requires broader validation across diverse real-world scenarios. Future Research Directions: Prioritizing transfer learning-based multi-source domain adaptation, expanding and diversifying datasets, and integrating mechanistic-empirical principles.

4. Conclusions

This study successfully developed and validated a hybrid (RF-DNN) framework, enhanced by a two-phase transfer learning strategy

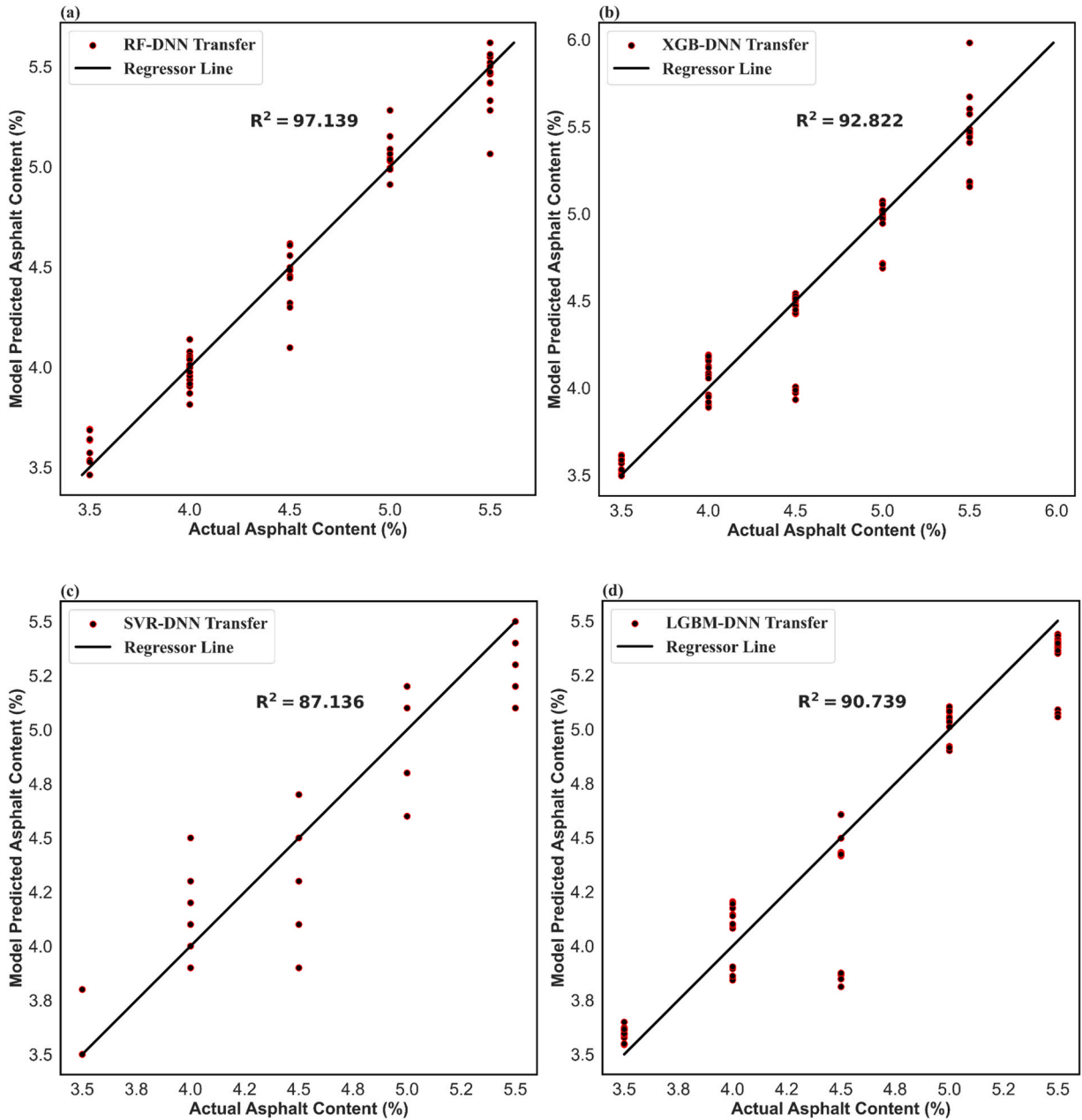


Fig. 16. Comparison of measured and predicted Asphalt Content using different hybrid models.

(RF-DNN-T), to significantly reduce the laboratory effort required for AC property prediction. The primary contribution of this research is the demonstration of a robust model that maintains high predictive accuracy even with limited target data, offering an efficient and scalable solution for intelligent pavement material design.

The proposed RF-DNN-T model demonstrated excellent accuracy and generalization, achieving a testing R^2 of 0.97 and the lowest prediction errors (MAE of 0.08, MSE of 0.02, and MAPE of 1.61) when compared to standalone RF, DNN, and other common machine learning approaches. This superior performance is attributed to the model's architecture, which synergizes the feature selection capabilities of RF with the advanced representation learning of DNN. Furthermore, the transfer learning strategy proved highly effective at mitigating domain shift, thereby enhancing the model's predictive accuracy and cross-domain generalization.

A key insight from the model, revealed through SHAP analysis, is the dominant influence of volumetric properties on AC performance. Air Voids (Va) were identified as the most critical feature, contributing approximately 32 % to the prediction, followed by Voids Filled with Asphalt (VFA) (22 %) and Voids in Mineral Aggregate (VMA) (17 %).

In conclusion, this work addresses the critical challenge of data scarcity in pavement engineering by providing a validated

framework that leverages pre-trained knowledge to supplement limited experimental datasets. This approach not only conserves time and resources but also establishes a strong foundation for applying pre-trained models to other complex material property studies. Future research should continue to explore transfer learning, particularly leveraging knowledge from multiple source domains, which presents a promising avenue for overcoming data limitations in real-world engineering applications.

CRedit authorship contribution statement

Ming-Der Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Funding acquisition, Conceptualization. **Yared Bitew Kebede:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Formal analysis, Conceptualization. **Henok Desalegn Shikur:** Visualization, Validation, Software, Formal analysis, Data curation.

Declaration of Competing Interest

Ming-Der Yang, hereby declare my interests in relation to the manuscript titled submitted to Case Studies in Construction Materials.

1. **Financial Interests:** I have no financial interests or affiliations with any individuals, organizations, or entities that could potentially influence the research or the publication of this paper.
2. **Non-Financial Interests:** I have no non-financial conflicts of interest, including but not limited to personal relationships, academic competition, or intellectual bias, that might affect the integrity, objectivity, or credibility of the research presented in this paper.
3. **Sources of Funding:** This research received no external funding, and I declare that there are no sources of financial support that could influence the content or outcomes of the study.
4. **Previous Collaborations:** I acknowledge that I have collaborated with some of the co-authors on previous research projects, but these collaborations did not involve financial arrangements or other interests that would compromise the validity or impartiality of the findings presented in this paper.
5. **Patent and Licensing:** I confirm that there are no patents or licensing agreements associated with the research presented in this paper that could create conflicts of interest.

I affirm that the above information is accurate and complete to the best of my knowledge. I am committed to upholding the highest standards of transparency and ethics in the publication of this manuscript.

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Data availability

Data will be made available on request.

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